

IEDA 4000H

Homework 1

Due Date: Sep 24 (Wednesday)

1. You are a CFA (chartered financial analyst). Betty has come to you because she needs help paying off her credit card bills. She owes the amounts on her credit cards shown in Table 1. Betty is willing to allocate up to \$6000 per month to pay off these credit cards. All cards must be paid off within 6 months. Betty's goal is to minimize the total of all her payments. To solve this problem, you must understand how interest on a loan works. To illustrate, suppose Betty pays \$6000 on BOCHK during month 1. Then her BOCHK balance at the beginning of month 2 is

$$10000 \times (1 + 0.005) - 6000 = 4050.$$

This follows because during month 1 Betty incurs $10000 \times 0.5\%$ in interest charges on her BOCHK card.

Table 1: Credit Card Bills

Bank	Balance(\$)	Monthly Rate(%)
BOCHK	10000	0.5
HSBC	15000	1.0
SC	5000	1.5

- (a) Formulate a linear program to minimize the total payments Betty makes over the next six months.
- (b) Solve the linear program from the previous part using the CVXPY library in Python.

2. Let $S_{++}^n = \{X \in \mathbb{R}^{n \times n} \mid X^\top = X, X \succ 0\}$.

(a) Show that S_{++}^n is a convex set.

(b) Consider the function $f(X) = \log \det X$ with $\text{dom} f = S_{++}^n$. Show that f is concave.

3. Let Ω be a nonempty set, convex and bounded subset of $\mathbb{R}^n$. Define

$$\mu_\Omega(\mathbf{x}) := \sup\{\|\mathbf{x} - \boldsymbol{\omega}\| \mid \boldsymbol{\omega} \in \Omega\}, \quad \mathbf{x} \in \mathbb{R}^n.$$

Show that

(a) μ_Ω is well defined, i.e. $\mu_\Omega(\mathbf{x}) < \infty$ for all $\mathbf{x} \in \mathbb{R}^n$.

(b) μ_Ω is convex.